Responsibly deploy AI in Quantitative Social Research

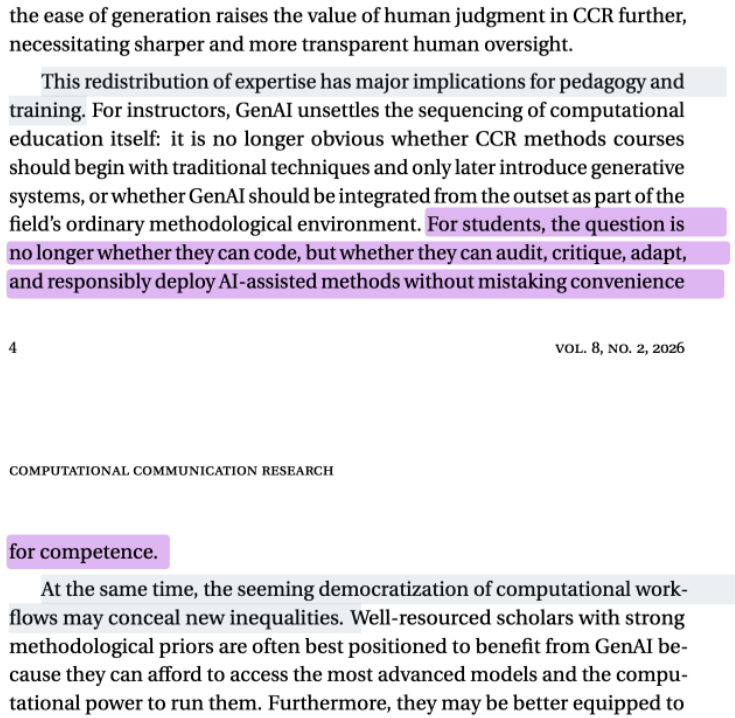
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Research Interest: Using causal and computational methods to evaluate societal and political impacts of digital information technologies

November 5, 2026

The goal of this session

... lower barrier (of computational skills) ... redistribution of expertise ... "For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience"



the ease of generation raises the value of human judgment in CCR further, necessitating sharper and more transparent human oversight.

This redistribution of expertise has major implications for pedagogy and training. For instructors, GenAI unsettles the sequencing of computational education itself: it is no longer obvious whether CCR methods courses should begin with traditional techniques and only later introduce generative systems, or whether GenAI should be integrated from the outset as part of the field's ordinary methodological environment. For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience

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COMPUTATIONAL COMMUNICATION RESEARCH

for competence.

At the same time, the seeming democratization of computational workflows may conceal new inequalities. Well-resourced scholars with strong methodological priors are often best positioned to benefit from GenAI because they can afford to access the most advanced models and the computational power to run them. Furthermore, they may be better equipped to

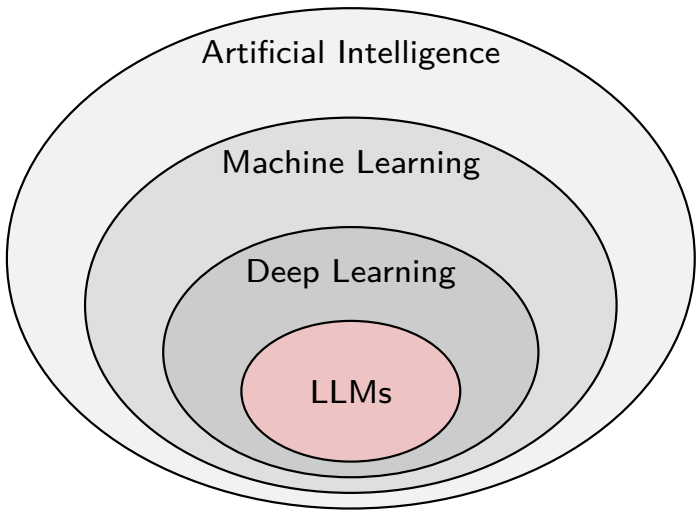
Source: **zhou2026be<empty citation>** (pp. 4–5)

Introduction

What Is AI?

- "The science and engineering of making intelligent machines"

  (McCarthy)
- Systems that **perceive** their environment and **act** to achieve goals
- A nested family of techniques, not a single technology



A Brief History of AI

- Observation: Artificial intelligence is a general field developed decades ago (singple-poatrone); Early tools mainly classified, clustered, or predicted from human-produced traces (zhou2026be).
- Roughly into pre-LLM and LLM eras for the goal of this session
  - Before the introduction of Transformers (vaswani2017attention), a model structure designed to parallelize training, we have traditional machine learning models such as logistic regression, SVMs, and early neural networks
  - After the introduction of Transformers, large-scale pre-trained language models such as BERT and GPT series became prominent, enabling more general-purpose natural language understanding and generation tasks, such as generative language models and conversational agents
  - Specifically, we focuse on the LLM era post-2020 (xu2026large) and its implications for computational social science research

Pre-LLMs AI

The use of Pre-LLM AI in Quantitative Social Research

- **Supervised learning**: hand-label a sample, train a classifier, scale up
- **Unsupervised learning**: topic models, clustering, word embeddings
- **Text, images, and networks as data** (king2013how)
- **ML for causal inference**: heterogeneous effects, high-dimensional controls

Common thread: task-specific models, labeled data, and a researcher-built pipeline

and computational linguistics, and later in the social sciences and digital humanities—developed tools, models, and software that facilitated the analysis and organization of texts at scale.

Almost all of the applications of large-scale text analysis in the social sciences use algorithms either first developed in computer science or built closely on those developments. For example, numerous papers within political science—including many of our own—build on topic models (Blei, Ng, and Jordan, 2003; Quinn et al., 2010; Grimmer, 2010; Roberts et al., 2013) or use supervised learning algorithms for document classification (Joachims, 1998; Jones, Wilkerson, and Baumgartner, 2009; Stewart and Zhukov, 2009; Pan and Chen, 2018; Barberá et al., 2021). Social scientists have also made methodological contributions themselves, and in this book we will showcase many of these new models designed to accomplish new types of tasks. Many of these contributions have even flowed from the social sciences to computer science. Statistical models used to analyze roll call votes, such as Item Response Theory models, are now used in several computer science articles (Clinton, Jackman, and Rivers, 2004; Gerrish and Blei, 2011; Nguyen et al., 2015). Social scientists have broadly adapted the tools and techniques of computer scientists to social science questions.

However, the knowledge transfer from computer science and related fields has created confusion in how text as data models are applied, how they are validated, and how their output is interpreted. This confusion emerges because tasks in academic computer science are different than the tasks in social science, the digital humanities, and even parts of industry. While computer scientists are often (but not exclusively!) interested in information retrieval,

Source: **grimmer2022text<empty citation>** (p.23)

LLMs

Roles of LLMs in Computational Social Science

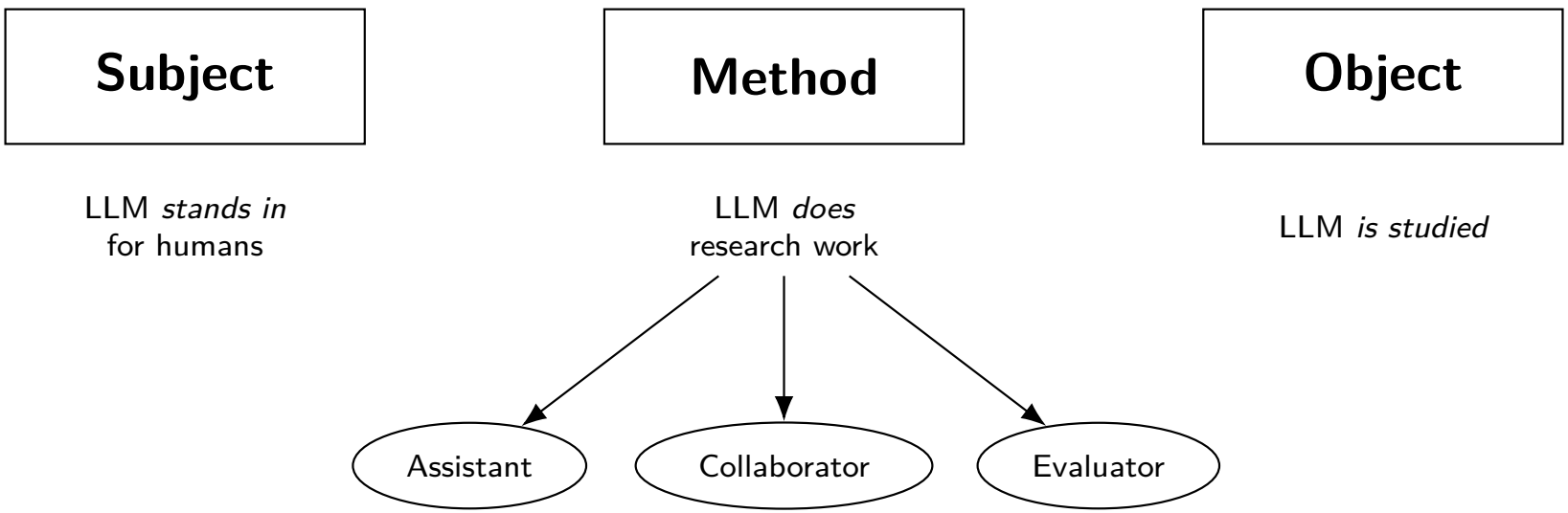
- A recent review article suggests four roles of LLMs in computational social science: Subject, Assistant, Collaborator, Evaluator

Table 1 | Representative recent work leveraging LLMs in social scientific study, organized by the research roles they have

Research role	Description	Example works
Research subjects	Generating survey responses	Refs. 32,43,48
	Simulating behaviour sequences	Refs. 52,53,55
	Simulating social interaction	Ref. 61,63,65,67
Research assistants	Interviewing human participants	Refs. 44,73,74,76
	Supporting experiments	Refs. 81–83
	Analysing textual content	Refs. 85,93,99,102
	Mining social networks	Refs. 107–109
Research collaborators	Reviewing literature	Refs. 117,120,122
	Proposing hypotheses	Refs. 126–128
	Designing research protocols	Refs. 129,130
Research evaluators	Evaluating research protocols	Refs. 132,133,137,140
	Critiquing research outcomes	Refs. 141,142,144

Roles of LLMs in Computational Social Science

- By adopting more recent perspectives, we may propose a more comprehensive categorization of LLM roles in CSS; to be elaborated in the following slides



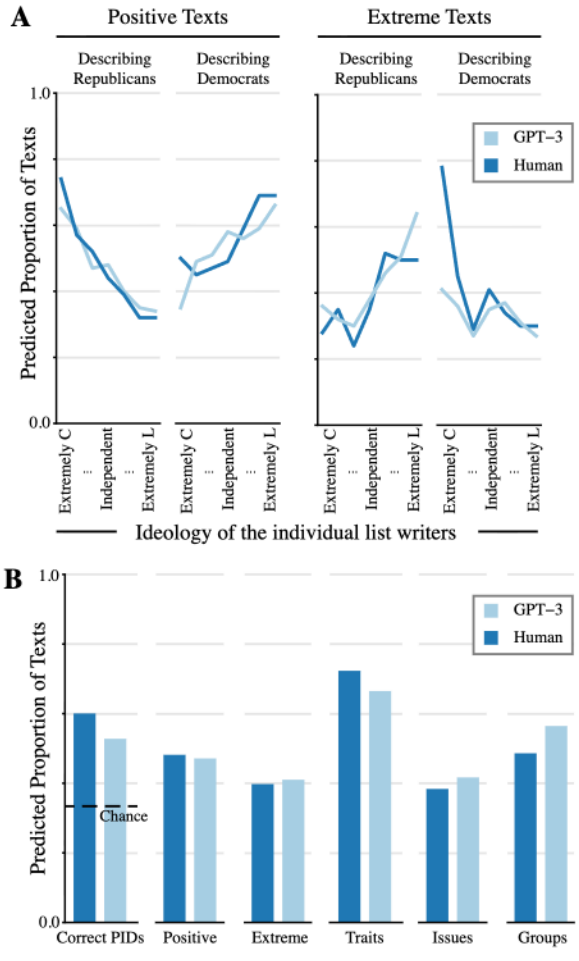
Source: Four roles (subjects, assistants, collaborators, evaluators) categorized in **xu2026large**<**empty citation**>; four categories thought from **liang2026computational**<**empty citation**>; and two roles (a research tool, an object of inquiry) adapted from **zhou2026be**<**empty citation**>

AI as Subject: Categories

- LLMs serve as research subjects
  - Generate survey responses
  - Simulating behavior sequences
  - Simulating social interactions

AI as Subject: Pros and Cons

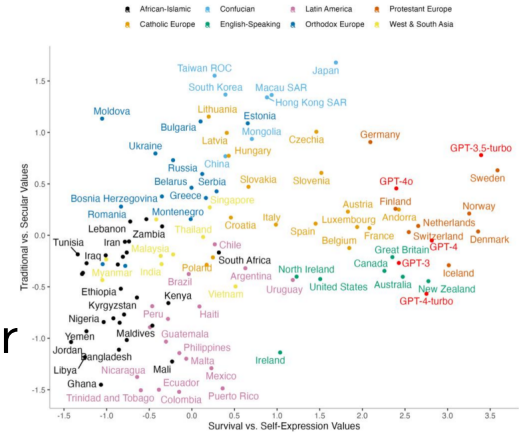
- **Pros:**
  - Successfully replicate human responses (argyle2023out)



Analysis of GPT-3 and human responses from the Lucid su; Fig 3 in argyle2023out<empty citation>

AI as Subject: Pros and Cons

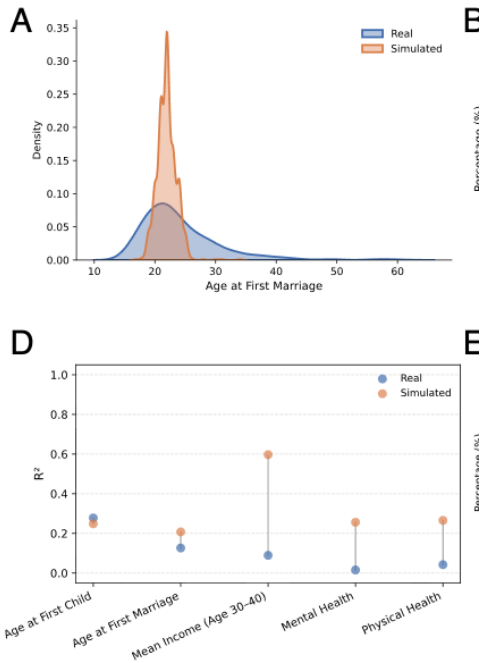
- **Cons:**
  - Poorly refelcting "hard-to-reach populations" (bollen2026nationally); fail to capture experiential and cultural depth (pataranutaporn2025simulating)
  - Reflect certain partisan (santurkar2023whose) or cultural biases (tao2024cultural)
  - Closed to averages but less variation (bisbee2024synthetica); have strong, varying biases and have low-variance in opinnion or attitudinal research(boelaert2025machine); likely to both misportray and flatten demographic representations (wang2025large); and tend to compress heterogeneity into simplified typological structures (xie2025variance)



Deviated LLM-generated answers related to values compared with different countries; Fig 1 in **tao2024cultural**<**empty citation**>

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Failing to capture the dispersion of real-world distribution; Fig 3 in xie2025variance<empty citation>

AI as Subject: Validation

- **Cons:**

LLM as Method: Assistant

- Writing, debugging, and translating analysis code
- Searching and summarizing literature
- Editing and polishing prose
- Data cleaning and wrangling

The researcher specifies the task; the LLM speeds up execution

LLM as Method: Collaborator

- Brainstorming research questions and hypotheses
- Critiquing research designs and identification strategies
- Proposing measures, robustness checks, and alternative explanations
- Agentic workflows: multi-step analysis with researcher oversight

The LLM contributes ideas; the researcher remains accountable

LLM as Method: Evaluator

- **Annotation**: zero- and few-shot text classification at scale
- **LLM-as-judge**: scoring quality, stance, or similarity
- **Review**:
- **Replication**: To scale up reproducibility of empirical work

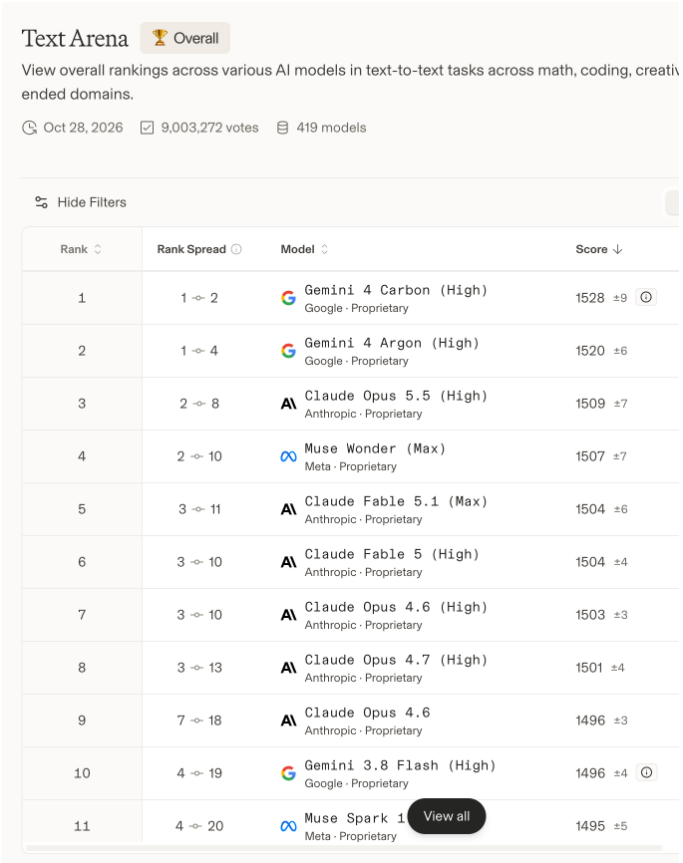
(xu2026scaling)

LLM as Method: Choose the Right Tool

- Consider a model as a function, $f(data, algo, object)$, the differences in each part would substantially shape their in-domain capabilities (like better coding or writing)
- In this sense, we may need to choose different models for different tasks, depending on their strengths and limitations

LLM as Method: Choose the Right Tool

- For example, you may expect Gemini Carbon to excel in writing tasks whereas Claude Opus 5.5 outperforms in engineering



Text Arena — Overall

View overall rankings across various AI models in text-to-text tasks across math, coding, creativ ended domains.

Oct 28, 2026 · 9,003,272 votes · 419 models

Hide Filters

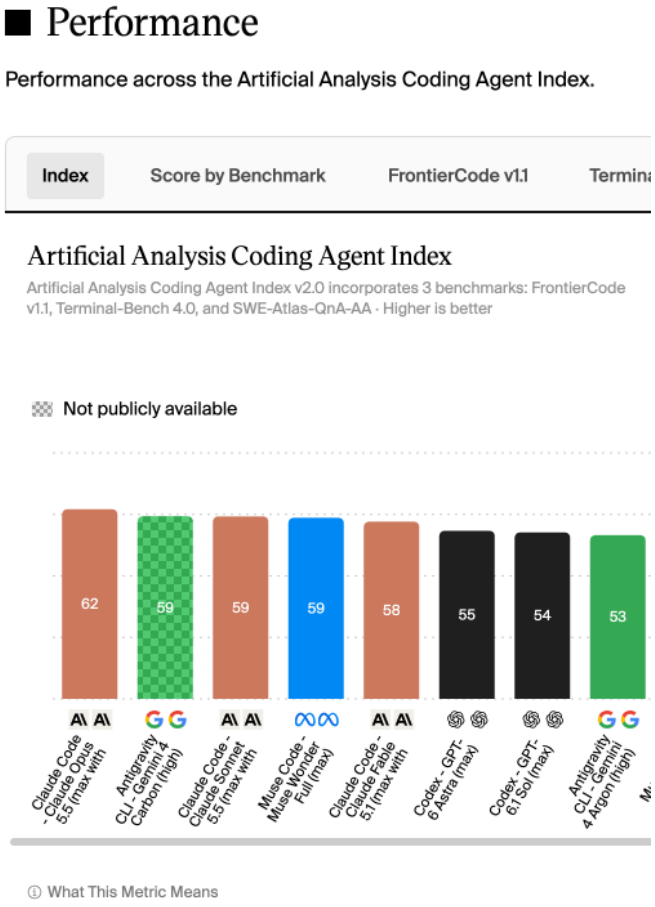
Rank	Rank Spread	Model	Score
1	1 → 2	Gemini 4 Carbon (High) — Google · Proprietary	1528 ±9
2	1 → 4	Gemini 4 Argon (High) — Google · Proprietary	1520 ±6
3	2 → 8	Claude Opus 5.5 (High) — Anthropic · Proprietary	1509 ±7
4	2 → 10	Muse Wonder (Max) — Meta · Proprietary	1507 ±7
5	3 → 11	Claude Fable 5.1 (Max) — Anthropic · Proprietary	1504 ±6
6	3 → 10	Claude Fable 5 (High) — Anthropic · Proprietary	1504 ±4
7	3 → 10	Claude Opus 4.6 (High) — Anthropic · Proprietary	1503 ±3
8	3 → 13	Claude Opus 4.7 (High) — Anthropic · Proprietary	1501 ±4
9	7 → 18	Claude Opus 4.6 — Anthropic · Proprietary	1496 ±3
10	4 → 19	Gemini 3.8 Flash (High) — Google · Proprietary	1496 ±4
11	4 → 20	Muse Spark 1 — Meta · Proprietary	1495 ±5

View all

Source: https://arena.ai/leaderboard/chat/text (Nov 3, 2026)

LLM as Method: Choose the Right Tool

- For example, you may expect Gemini Carbon to excel in writing tasks whereas Claude Opus 5.5 outperforms in engineering
- In general, model labs have different commercialization models, for business model optimized for engineering tasks may prioritize coding capabilities (e.g., Anthropic); please check benchmarks / review the literature / do a pilot test before using a model



Source: https://artificialanalysis.ai/agents/coding-agents#coding-agents-index (Nov 3, 2026)

LLM as Object

- **Auditing models**: political slant, social bias, and value alignment
- **Effects on people**: persuasion, learning, labor, and well-being
- **Effects on information ecosystems**: AI-generated content and search

Limitations

Limitations

1. **Validity.** Outputs may not measure what we intend
2. **Reproducibility.** Closed models, silent updates, and stochastic outputs
3. **Bias.** Training data over-represent some groups and languages
4. **Hallucination.** Fabricated facts and citations
5. **Ethics.** Privacy, consent, and data sent to third parties
6. **Over-reliance.** Deskilling and unexamined defaults

Takeaways

Takeaways

- AI models (which includes LLMs) can be subjects, objects, and methods in quantitative social research
- As methods, they are expected to be used in a responsible way in which validity, reproducibility, bias, hallucination, ethics, and over-reliance are carefully considered
- Like any other methods, they evolve in a quick (and potentially quicker pace), but our tastes of good questions remain relatively central to research (and industrial jobs)
- "... Be skeptic... don't panic!" (angrist2009mostly)

Takeaways

- "... Be skeptic... don't panic!" (angrist2009mostly)

Last words

AI Social ICML

If applied econometrics was easy, theorists would do it. But it's not as hard as the dense pages of *Econometrica* might lead you to believe. Carefully applied to coherent causal questions, regression and 2SLS almost always make sense. Your standard errors probably won't be quite right, but they rarely are. Avoid embarrassment by being your own best skeptic - and, especially, Don't Panic!

AI

Source: angrist2009mostly<empty citation>

Resources

Resources

- Locate the most appropriate AI model for your task
  - Artificial analysis https://artificialanalysis.ai/
  - Arena.ai, blind benchmarking for AI models https://arena.ai/leaderboard/
- Find problems in your manuscript
  - Refine, publication-level technical review; adopted by AEA https://www.refine.ink/
  - OpenAIReview, opensource solution of multi-agent manuscript review by UChicago HAI https://github.com/ChicagoHAI/OpenAIReview
- Doublecheck references and citations
  - CiteVerify, "a local tool for verifying references in research papers", https://github.com/cuihuash/CiteVerify
- Manage multiple AI models and services
  - For macOS, magpie, local AI router management for mac https://github.com/yetone/magpie
  - For other platforms, cc-switch, https://github.com/farion1231/cc-switch
- Read scientific literature using your own language

References

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